CFD-trained machine learning surrogate for aerodynamic drag prediction and shape optimization

By

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Abstract

This project built and validated a machine learning surrogate model for aerodynamic drag prediction, trained entirely on CFD data generated by an automated pipeline. A simplified streamlined vehicle body was controlled by five shape parameters while the frontal area was kept fixed. 120 design variations were selected using Latin Hypercube sampling and simulated in OpenFOAM (steady state incompressible, k-omega SST, 30 m/s) through a fully automated Docker pipeline, completing all 120 runs in 14.9 hours on a single laptop with zero failures. A mesh independence study was performed first, showing that the coarse mesh over-predicted drag by 18%, so a finer mesh accurate to within 0.6% of a reference mesh was used for the campaign. A Gaussian Process regression model trained on the results predicted the Cd of unseen designs with $R^2 = 0.981$ and an average error of 0.005. The surrogate was then used as the scoring function for OpenEvolve, which found an optimized shape in 200 iterations. A final blind CFD run confirmed the result: the surrogate predicted Cd = 0.0493 and CFD gave Cd = 0.0470, within the model's own uncertainty estimate. The optimized body achieved 64% lower drag than the baseline seed design and 20% lower drag than any design in the training data, validating the complete data-to-design workflow.

1. Objective and Introduction

Aerodynamic drag directly affects vehicle efficiency, so finding low drag shapes is a core design task. The problem is that evaluating each candidate shape with CFD takes minutes to hours, which makes it impossible to explore a large design space by simulation alone. The industry solution is a surrogate model: run a limited, well chosen set of CFD simulations once, train a fast prediction model on that data, and let an optimizer query the model instead of the solver. This is the same approach used in professional aerodynamic development, including Formula 1 and automotive programs.

In the previous car body project, the surrogate used by OpenEvolve was a hand-written physics approximation inspired by Ahmed body behavior. The objective of this project was to replace that hand-written model with a real machine-learned surrogate trained on CFD data generated from scratch, and to validate every step of the chain. The project set out to answer four questions. First, whether an automated CFD pipeline could produce a trustworthy 120-case dataset, with mesh quality proven by a convergence study rather than assumed. Second, how accurately a model trained on roughly 100 samples could predict the drag of shapes it had never seen. Third, whether OpenEvolve, using only the surrogate as its judge, could find a design better than anything in the training data. Fourth, whether that design would hold up in a final blind CFD validation run.

2. Geometry and Design Parameters

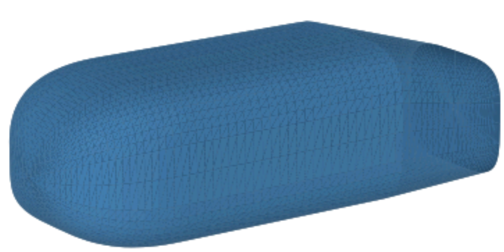

Figure 1. Baseline (seed) geometry: mid-range values of all five design parameters.

The geometry is a deliberately generic streamlined body with vehicle-like proportions: a rounded nose, a constant mid-section, and a tapered rear ending in a flat vertical base (Kammback style), positioned 5 cm above the ground. It was generated purely in Python as a watertight STL from cross-sections stacked along the body length. Five parameters control the shape, while the overall length, width, and height are locked so that every design shares the same frontal area of 0.1048 m^2. This ensures that any difference in drag coefficient comes from shape changes rather than size changes, consistent with the approach used in the previous car body project. The backlight angle range deliberately includes the region associated with the Ahmed body drag crisis so that the dataset contains genuinely nonlinear aerodynamic behavior.

Parameter	**Baseline value**	**Allowed range**
Body length	1.00 m	fixed
Maximum width	0.39 m	fixed
Maximum height	0.29 m	fixed
Frontal area	0.1048 m^2	fixed
Nose length fraction	0.25	0.15 – 0.35
Nose bluntness exponent	2.5	1.5 – 4.0
Tail length fraction	0.28	0.15 – 0.40
Backlight angle	15.0 degrees	5 – 30 degrees
Boat-tail angle	10.0 degrees	0 – 20 degrees

Table 1.

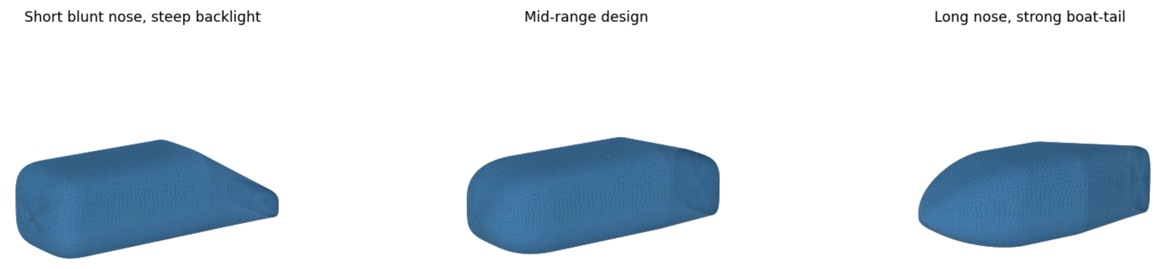

Figure 2. Three example designs showing the variety of shapes the five parameters can produce.

3. Design of Experiments and Automated CFD Pipeline

120 designs were selected with Latin Hypercube sampling, which spreads the samples evenly through the five-dimensional design space and gives far better coverage than a grid for the same number of runs. A fixed random seed makes the dataset reproducible. Each design was exported as an STL and simulated by an automated pipeline: for every case, the pipeline builds a complete OpenFOAM case, meshes the geometry with snappyHexMesh, runs a steady state RANS solution with simpleFoam, and extracts the drag and lift coefficients averaged over the final 30% of iterations. The pipeline runs OpenFOAM inside Docker, so the entire campaign ran unattended on a single laptop. The initial plan was to run the campaign through the SimScale API, but the required LBM solver needed GPU quota not included in the academic plan, so the pipeline was rebuilt around containerized OpenFOAM instead. All 120 simulations completed in 14.9 hours (7.4 minutes average per case) with no failed runs.

CFD setting	**Value**
Software	OpenFOAM v2412 (Docker container)
Solver	simpleFoam, steady state incompressible
Turbulence model	k-omega SST with wall functions
Inlet velocity	30 m/s (Re = 2×10^6 based on body length)
Fluid	Air, $\nu = 1.5\times10^{-5}$ m²/s, $\rho = 1.225$ kg/m³
Domain size	9 m × 4 m × 2.55 m (approx. 1% blockage)
Inlet / outlet	Velocity inlet / pressure outlet (0 Pa gauge)
Ground	No-slip wall, 5 cm below the body
Sides and top	Slip walls
Body surface	No-slip wall
Mesh	snappyHexMesh, refinement box around body and wake
Iterations	1000, coefficients averaged over final 30%

Table 2.

4. Mesh Independence Study

Before committing to 120 runs, the same design (run_000) was simulated at three mesh refinement levels. The coarse mesh (134k cells) over-predicted Cd by 18% relative to the fine mesh, which would have contaminated the entire dataset with mesh error. The fine mesh agreed with a much finer reference mesh to within 0.6%, so the fine mesh was adopted as the production mesh for the campaign. This step converted the mesh quality from an assumption into a measured result.

Mesh level	Cd (run_000)	Difference vs fine
Coarse (134k cells)	0.1223	+18.0%
Fine (production)	0.1036	—
Very fine (reference)	0.1042	+0.6%

Table 3.

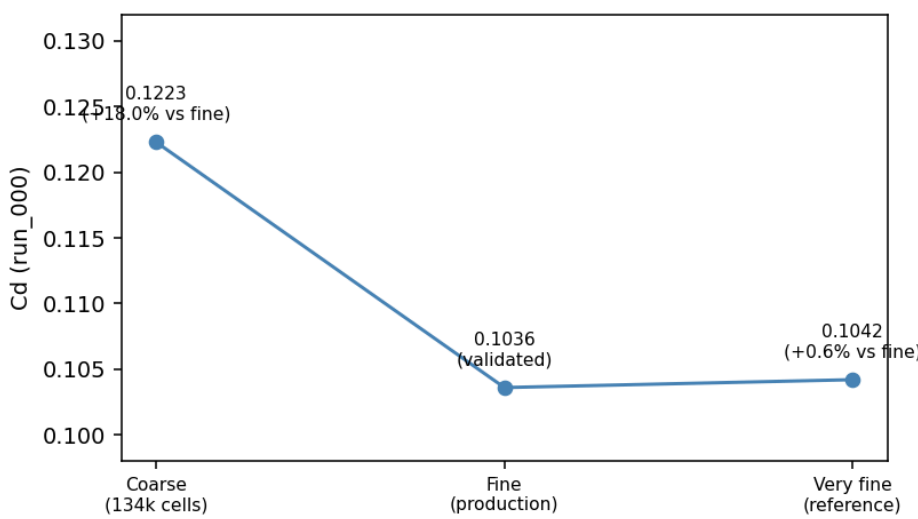

Figure 3. Mesh convergence for design run_000. Cd stops changing between the fine and very fine meshes.

5. CFD Dataset

The campaign produced a complete 120-row dataset with the five design parameters as inputs and Cd, Cl, drag force, and lift force as outputs. Drag coefficients ranged from 0.0587 to 0.2624 with a mean of 0.151, a factor of more than four between the best and worst design, confirming that the five parameters produce a wide range of aerodynamic behavior. Lift coefficients ranged from −0.218 to +0.305.

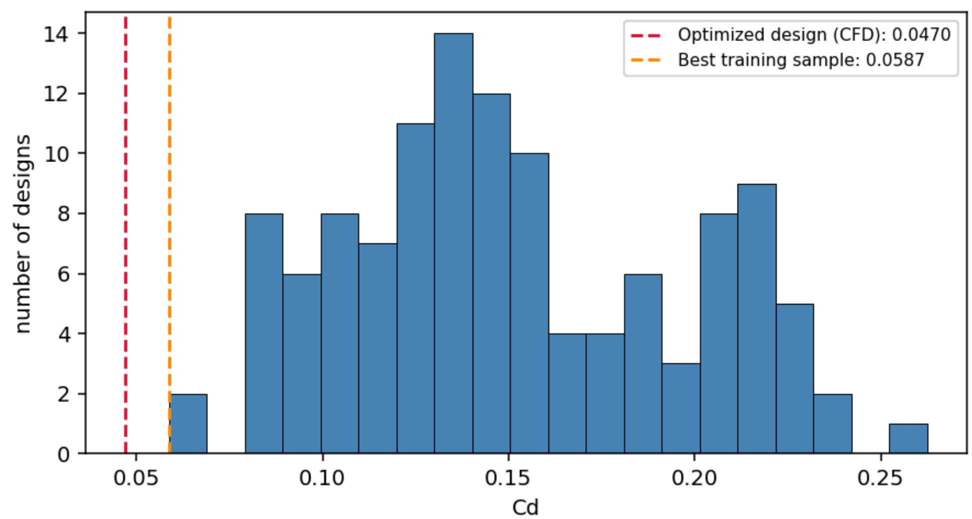

Figure 4. Distribution of drag coefficients across the 120 CFD simulations. The optimized design found later (red dashed line) lies below the entire training distribution.

6. Surrogate Model Training and Validation

Two model families were trained to predict Cd and Cl from the five shape parameters: Gaussian Process regression (GPR) and gradient boosting. The data was split into 96 training designs and 24 test designs that the models never saw during training. GPR clearly outperformed gradient boosting on this dataset size, predicting the drag coefficient of unseen designs with R^2 =

0.981 and a mean absolute error of 0.0052, which is comparable to the mesh uncertainty of the CFD itself. This matches expectations: Gaussian Processes are well suited to small engineering datasets, while tree-based models typically need more data. A further advantage of GPR is that every prediction comes with an uncertainty estimate, which is used directly in the optimization step.

Target	**Model**	**5-fold CV R^2**	**Test R^2**	**Test MAE**
Cd	GPR	0.965	0.981	0.0052
Cd	Gradient boosting	0.907	0.886	0.0125
Cl	GPR	0.994	0.997	0.0040
Cl	Gradient boosting	0.937	0.956	0.0175

Table 4.

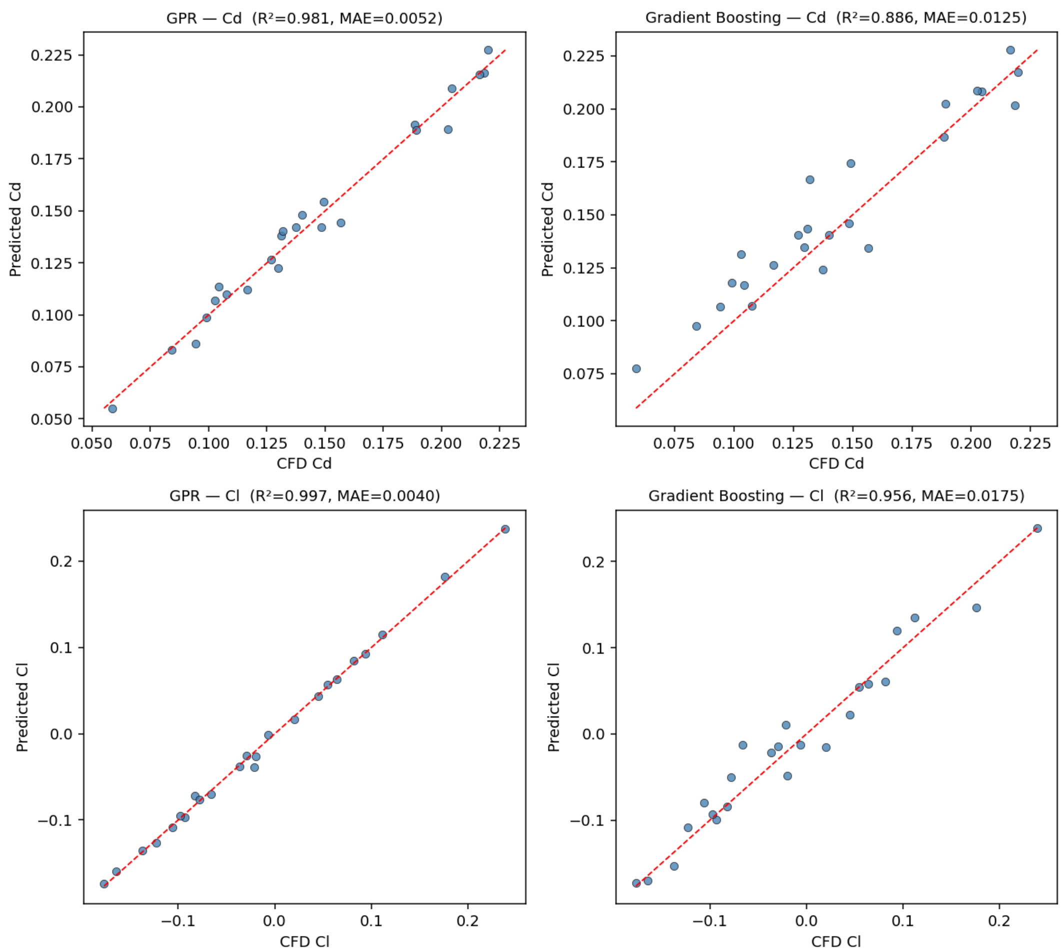

Figure 5. Predicted versus CFD values on the 24 held-out test designs. Points on the dashed line are perfect predictions.

A permutation importance analysis of the GPR model showed that Cd is dominated by the boat-tail angle, followed by the tail length, with the nose parameters contributing far less. This agrees with the physics of streamlined ground vehicles: drag is controlled mainly by the wake behind the rear base, which the rear taper parameters directly shrink, while nose details matter little once the nose is reasonably rounded. The model learned this behavior purely from the data.

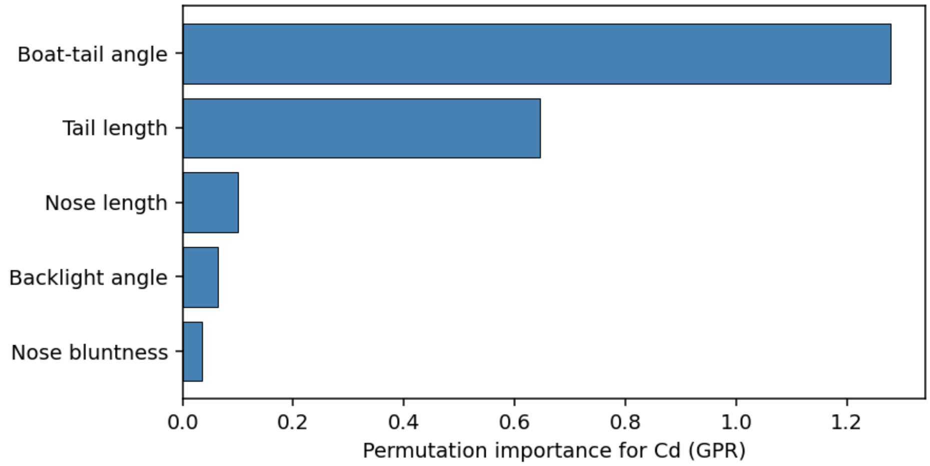

Figure 6. Permutation importance of each design parameter for Cd prediction.

Before optimization, the surrogate was tested blind on a manually chosen design outside the training set (long nose, near-maximum tail length, strong boat-tail). The model predicted Cd = 0.0549 ± 0.0057 and a dedicated CFD run gave Cd = 0.0589, an error of 6.8% and within the model's own uncertainty band. Notably, this prediction was below the best value in the training data, an early sign that the model generalizes rather than memorizes.

7. Optimization Method

The trained surrogate replaced the hand-written physics model in the standard OpenEvolve setup used in previous projects. OpenEvolve evolves only the five parameter values in a minimal seed file; a separate evaluator script, which the LLM never modifies, clamps every candidate to the training bounds and scores it with the surrogate. The score is the negative of the predicted Cd plus the model's uncertainty: score = −(Cd + σ). The uncertainty penalty is important — it prevents the optimizer from exploiting regions of the design space where the model has little data and its predictions cannot be trusted. Because each evaluation takes about a millisecond instead of 7.4 minutes, OpenEvolve could search the design space at a cost of roughly one hundred-thousandth of direct CFD per candidate. The run used gemini-2.5-flash for 200 iterations.

8. Optimized Geometry and CFD Validation

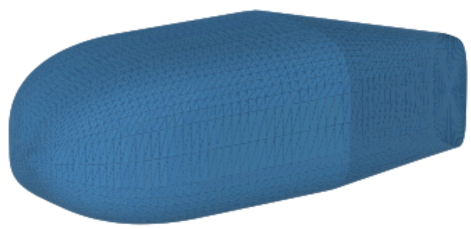

Figure 8. Optimized geometry found at iteration 192: long tail with strong boat-tailing and a moderate backlight angle.

The best design pushed the tail length and boat-tail angle to near their upper bounds, exactly the two parameters the importance analysis identified as dominant, while settling on a moderate 11.8 degree backlight angle rather than the maximum — correctly avoiding the steep-slant high-drag behavior seen in Ahmed body experiments. The optimizer was never told any of this physics; it emerged from the data through the surrogate.

Parameter	Baseline	Optimized	Change
Nose length fraction	0.25	0.305	+22.0%
Nose bluntness exponent	2.5	2.3	−8.0%
Tail length fraction	0.28	0.395	+41.1%
Backlight angle	15.0 deg	11.8 deg	−21.3%
Boat-tail angle	10.0 deg	19.5 deg	+95.0%
Surrogate Cd	0.1323	0.0493	−62.7%

Table 5.

The final and most important test was a blind CFD validation of the optimized shape. The surrogate predicted Cd = 0.0493 ± 0.0057; the CFD run gave Cd = 0.0470, an error of 0.0024 (5.1%), inside the model's stated uncertainty. The lift prediction was similarly accurate (−0.176 predicted versus −0.179 from CFD). The optimized design is 64% lower in drag than the baseline seed and 20% lower than the best of the 120 training designs — the optimizer found a better shape than any it had ever seen data for, and real CFD confirmed it.

Metric	Surrogate prediction	CFD result	Error
Cd	0.0493 ± 0.0057	0.0470	0.0024 (5.1%)
Cl	−0.176	−0.179	0.003

Table 6.

9. Limitations

This project has several limitations worth stating. The geometry family is simplified and does not include real-vehicle features such as wheels, an underbody, or detailed surface curvature, so the results describe shape trends rather than a production vehicle. The mesh independence study was performed on a single design (run_000); mesh error could differ at other points in the design space, especially at extreme tail angles. The mesh used no prism layers, which under-resolves wall friction, meaning absolute Cd values are likely somewhat low even though comparisons between designs remain valid. The surrogate was validated blind against CFD at only two points, both successful but limited in number. Finally, all data corresponds to a single operating condition of 30 m/s at zero yaw using steady RANS. Each of these limitations is addressable with additional computing time: mesh verification at the corners of the design space, prism layer meshing, additional validation runs, and datasets spanning multiple speeds and yaw angles.

10. Conclusion

This project demonstrated the complete surrogate-assisted design loop used in professional aerodynamic development, built and validated end to end: automated CFD data generation (120 runs, zero failures, mesh independence proven), machine learning surrogate training with honest held-out validation ($R^2 = 0.981$ for Cd), evolutionary optimization guarded by the model's own uncertainty estimate, and final blind CFD verification. The optimized body achieved a CFD-confirmed Cd of 0.0470 — 64% below the baseline and 20% below every design in the training data — with the surrogate's prediction landing within its stated error band in both blind tests. The key conclusion is that a properly validated surrogate does not merely interpolate its training data: it generalizes well enough to guide an optimizer into unexplored regions of the design space and be proven right when checked against real physics.